## Data visualization:

Raw Data → visual context.

↓

Hidden Patterns.

✓ TRENDS.

Relationships.

Data → Machine → Insights

↓

Visuals ⇒ PowerBI

## Python:

(i) coding

↓

Script.

Python & R.

↓

✓ Dis: Expertise ✓

Mobtime

(ii) GUI.

↓

Drag and Drop.

PowerBI, Tableau ✓

easy. NO Skill.

less Time.

↓

Dashboard Reports.

## NUMPY:

↓

PANDAS. ⇒ Pd.Class. Function.

↓

Data Manipulation

Data Analysis.

df.describe().

↓

min — ?

max — ?

✓ Mean —

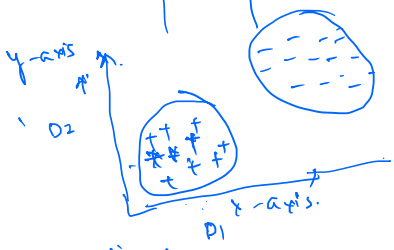
Median (Q2)

Q1

Q3

Diagnosis → +ve / -ve.

| Diagnosis | D1 | D2 |
|---|---|---|
| +ve | | |
| -ve. | | |




Machine Learning

## NUMPY:

↓

Matplotlib: ⇒ Foundation.

✓ Seaborn.

✓ Bokeh.

✓ Plotly and Dash.

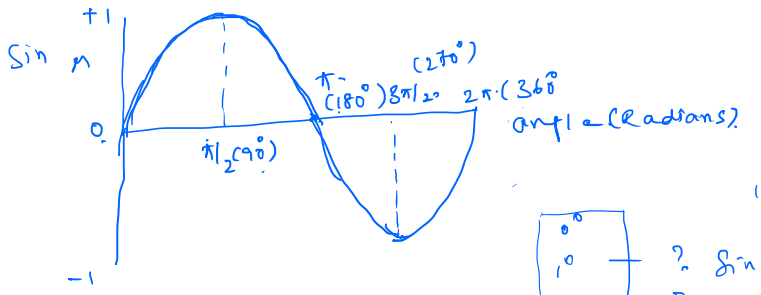
✓ Data Science Experts.

✓ Data Analyst.

✓ Mathematicians.

## Trigonometry:

Sin, cos, Tan.




| | |
|---|---|
| 0° | |
| 10° | ? |
| 20° | ? |
| 30° | ? |
| ⋮ | |
| 180° | ? |

15°

Sin(45° - 30°)

Sin 0 — 0

30 - 0.5 - (1/2)

45° - 0.707 - 1/√2)

60° - 0.866 (√3/2)

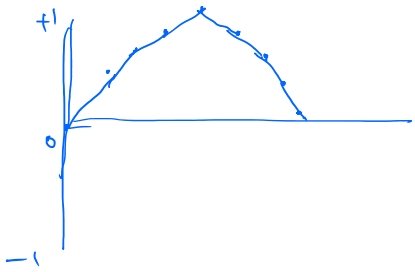
90° - 1 C

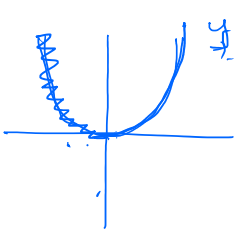
120° = 0.866 (√3/2)

135° = 0.707

150 = 0.5

180 = 0.



$y = (x^2)$